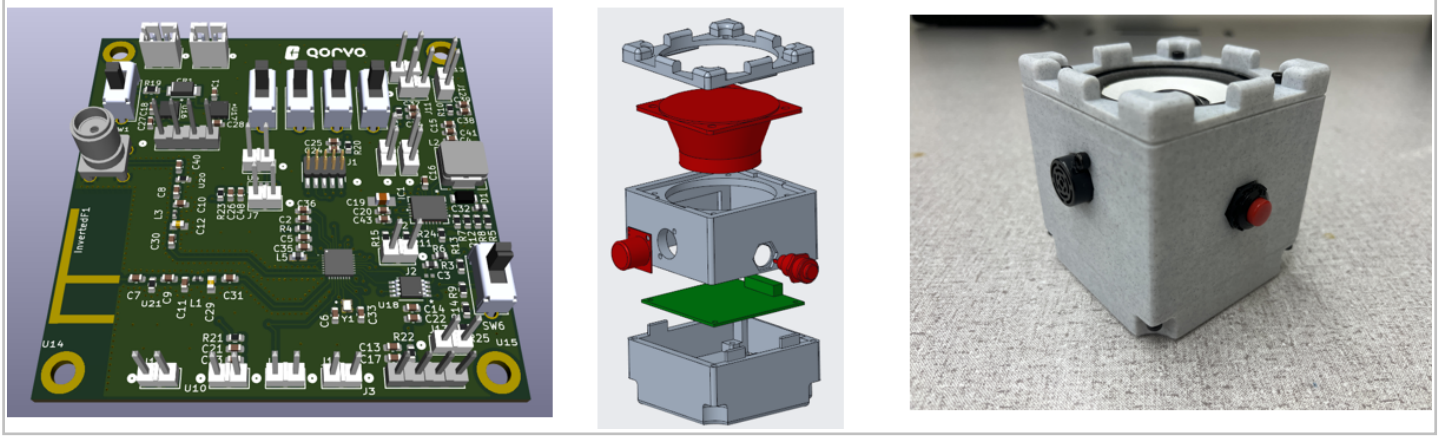
Soc Entryway Demonstration

QORVO_RADIO2



Images of PCB, exploded enclosure view, and actual enclosure

Team Members

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Mentors

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Project Overview

Our project's goal was to create devices using the Qorvo QPG6002L system on chip (SOC) that show off multi-protocol functionality and RF applications of the chip. To create these devices, we needed to create custom circuit boards, software, and mechanical enclosures. The devices we created were based around a smart doorbell and contain a doorbell, speaker, microphone, and sonic sensor to detect movement. These components along with the PCB are housed in a 3-d Printed PLA enclosure.

The devices are then connected to a gateway system and user interface. The user interface allows someone to see if a doorbell is being pushed or interact through the speaker or microphone. This is all connected to a smartphone or computer through Bluetooth or Thread.

In addition to showing of the capabilities of the SoC this project also had the purpose of promoting RF education and research at UNC Charlotte.